

Defective Analysis Addendum

Global Definitions, Operator Cores, and Repository Discipline

Written by N. Michael Kleban & Aleph

Postulated analysis needing confirmation and review

June 2026

Addendum to PDFs 1–4. This draft records framework objects that are adjacent to, or presupposed by, the four public Defective Analysis documents, but that have not yet been isolated as a single addendum.

Claim boundary. This addendum states proposed internal definitions and document-control rules. It uses ordinary mathematical language: set, class, map, quotient, fiber, kernel, section, residual, reconstruction, equivalence, admissibility, grammar, and extraction record. The symbols in this document are local notation for review. No claim is made that the framework is complete, correct, novel, universal, or applicable outside the declared formal setting. The purpose is to prevent drift between the public DA spine and the surrounding notes.

Abstract. The four public Defective Analysis documents fix the current spine: problem framing, Defective Calculus, a graph worked example, and COTG derived through the WDS path. This addendum records nine missing or underrepresented pieces: WDS as a standalone primitive, the Primal/Scalar Calculus hierarchy, Reconstruction Calculus as the operator core, the original COTG object, generator seeds, the clock-coordinate doctrine, the clock-order residual, retained-kernel admissibility, and thread/glossary discipline. The result is not a new theorem. It is a source-index layer for keeping future drafts typed, reviewable, and non-leaking.

1 Purpose and status

Definition 1 (Addendum role). The addendum role is to record objects that should be available before a reader attempts to merge the DA documents into one larger grammar. It is not a replacement for the four PDFs. It is a compatibility document.

The existing public sequence fixes the following spine:

$$DA \longrightarrow DC \longrightarrow \text{worked examples} \longrightarrow WDS \longrightarrow COTG.$$

This addendum records the surrounding definitions that make the spine easier to audit:

$$\text{seeds} \longrightarrow WDS \longrightarrow PC \longrightarrow SC, \quad \text{Recon} \subseteq DC \subseteq DA.$$

Definition 2 (Document admissibility). A DA document is document-admissible when every term it uses is either defined in that document, imported from a named upstream document, or marked provisional with an explicit proof obligation.

2 Whole Defect Systems as a standalone primitive

Definition 3 (Whole Defect System). A Whole Defect System is a tuple

$$N = (\mathcal{A}, X, \Delta, \{K_p\}_{p \in P}, G, q, \text{Adm}_N)$$

where:

- $\mathcal{A}$ is the declared ambient category or comparison setting;
- X is the carrier of states before quotient;
- Δ is the defect-extraction operator;
- K_p is a defect sector indexed by an internal label $p \in P$;
- G is the gluing or comparison carrier;

- q is the quotient or gluing projection identifying defect directions invisible at the declared resolution;
- Adm_N is the admissibility predicate for this WDS presentation.

The shorter tuple

$$N = (X, \Delta, \{K_p\}_{p \in P}, G, q)$$

is admissible as notation only when the ambient setting and admissibility predicate are fixed by context.

Definition 4 (Defect object). The defect object of N is the extracted data

$$\Delta N.$$

It is not necessarily a numerical error. It is the obstruction, distinguishability data, or sector information exposed by the declared extractor.

Definition 5 (Sector decomposition). A WDS admits a sector decomposition when its defect object has a declared decomposition

$$\Delta N \simeq \bigoplus_{p \in P} K_p.$$

When a binary sector is distinguished, write

$$\Delta N \simeq K_2 \oplus K_{\text{odd}}, \quad K_{\text{odd}} = \bigoplus_{p \in P, p > 2} K_p.$$

The notation $p > 2$ is internal sector grammar unless an external ordered index set has been separately constructed.

Remark 1. A WDS is upstream of COTG. It contains the carrier, extractor, sector data, comparison carrier, and quotient map. COTG is not the whole WDS; it is a realization or quotient layer admitted after the relevant no-leakage checks.

3 Primal Calculus and Scalar Calculus hierarchy

Definition 6 (Primal Calculus). Primal Calculus is the partial operator calculus paired with WDS and COTG data. A Primal Calculus instance consists of candidate systems, WDS presentations, COTG objects, residual generators, and partial reconstruction operators:

$$\text{PC} = (\text{Sys}, \text{WDS}, \text{COTG}, \text{Gen}_P, D_P, D_0, I_P, I_0, K_{\text{ret}}, \text{Adm}_P).$$

The operators are partial. A candidate system is not primal-admissible merely because it has been named.

Definition 7 (Primal derivative). For a WDS presentation N , the Primal derivative is defect extraction:

$$D_P(N) = \Delta N.$$

For a candidate system S , $D_P(S)$ is defined only if S is supplied with a WDS presentation and comparison data. In complete form one records

$$D_P(S) = (C_S, K_S) \in \text{Gen}_P,$$

where C_S is the admitted COTG object and K_S is a typed residual or retained generator over it.

Definition 8 (Primal integral). The complete Primal integral is partial reconstruction from a generator pair:

$$I_P(C, K) \longrightarrow S / \sim_S.$$

The flat integral omits the retained generator:

$$I_0(C) \longrightarrow S / (\sim_S \vee K_{\text{hid}}).$$

Thus I_0 reconstructs only modulo the hidden sector, while I_P may reconstruct up to the declared system equivalence when K is admissible.

Definition 9 (Scalar Calculus). Scalar Calculus is the scale-indexed sector of the framework. For a scale or comparison index s , a Scalar Calculus instance declares

$$\text{SC}_s = (X_s, \sim_s, K_s, \Delta_s, P_s, \text{COTG}_s, K_{s,\text{ret}}, I_s, \text{Adm}_s).$$

The intended grammar is

$$D_s(X) = \text{WDS}_s(X), \quad \text{COTG}_s(X) = K_s / K_{s,\text{hid}}, \quad I_s(X) = (\text{COTG}_s(X), K_{s,\text{ret}}(X)).$$

Scalar admissibility requires reconstruction compatibility:

$$\text{Recon}_s(\text{COTG}_s(X), K_{s,\text{ret}}(X)) \sim_s X.$$

Scalar Calculus is not a replacement for Primal Calculus. It is the finite or scale-indexed test layer in which quotient, retained kernel, and no-leakage claims can be made concrete.

4 Reconstruction Calculus as the operator core

Definition 10 (Reconstruction Calculus instance). A Reconstruction Calculus instance is a tuple

$$R = (X, \sim, Y, K, D, I, \Pi, \text{Res}, \text{Adm})$$

where $D : X \rightarrow Y$ is a defective observable, $I : Y \times K \rightarrow X / \sim$ is a reconstruction operation, Π is a visible quotient or comparison map, Res is a declared residual carrier or residual map, and Adm is an admissibility predicate.

Definition 11 (Flat and complete reconstruction). If $0 \in K$ is a declared null retained datum, then flat reconstruction is

$$I^0 D(X) = I(DX, 0).$$

Complete reconstruction is any reconstruction of the form

$$I(DX, k), \quad k \in K_{\text{ret}}(X).$$

The central reconstruction test is

$$I(DX, k) \sim X.$$

Definition 12 (Reconstruction equivalence). Two objects $X, X' \in X$ are reconstruction-equivalent relative to R when they have the same observable and admit the same reconstruction class after admissible retained data are supplied:

$$X \equiv_R X' \iff DX = DX' \text{ and } \exists k, k' \in K [\text{Adm}(X, k) \wedge \text{Adm}(X', k') \wedge I(DX, k) \sim I(DX', k')].$$

A stricter realization may replace existence by a specified section or by equality in a declared quotient object.

Proposition 1 (Containment). Reconstruction Calculus is the operator core of Defective Calculus. Defective Calculus adds clock labels, polarity, static/operational regime separation, and no-leakage discipline:

$$\text{Recon} \subseteq \text{DC} \subseteq \text{DA}.$$

This is a containment of grammar, not an assertion about subfields of standard mathematics.

5 Original COTG object

Definition 13 (Original COTG skeleton). The original COTG skeleton is the minimal parity-stable cyclic gluing object

$$\text{COTG}(N) = (A, B, C; E; K_2; \text{glue}_2),$$

where A, B, C are comparison sectors, E is an equality carrier, K_2 is the surviving parity kernel, and glue_2 is the binary gluing law.

Definition 14 (WDS-to-COTG quotient). If N is a WDS with decomposition

$$\Delta N \simeq K_2 \oplus K_{\text{odd}}$$

and quotienting by K_{odd} preserves the K_2 component, then the parity-stable quotient route is

$$\text{COTG}(N) \simeq N/K_{\text{odd}}$$

in the declared quotient category.

Definition 15 (Analytic COTG realization). A finite analytic COTG realization may be recorded as

$$C = (M, \mathcal{U}, E, \{X_i\}_{i \in I}, \{q_i\}_{i \in I}, \{g_{ij}\}_{(i,j) \in E}, T, \Pi, R),$$

where M is the base or index domain, $\mathcal{U}$ is a chart family, E is an overlap or gluing graph, X_i are local fibers, q_i are local state sections, g_{ij} are transition-like maps, T is a global tensor or replacement object, Π is a projection, and R is residual data.

Remark 2. The original COTG skeleton may be used as provisional notation only when its equality carrier, parity kernel, and gluing law are supplied. The matured DA ordering is WDS first, defect extraction second, parity quotient third, COTG realization fourth.

6 Generator seeds and seed gauge

Definition 16 (Seed gauge). A seed gauge is a declared primitive generator presentation from which later objects are generated or reconstructed. The two basic seed gauges currently distinguished are

$$G_{01} = \{0, 1\}, \quad G_{012} = \{0, 1, 2\}.$$

The first treats 2 as generated by $1 + 1$. The second treats 2 as present at the seed level.

Definition 17 (Generated two). Relative to G_{01} , the symbol 2 is generated by the successor operation:

$$2 = 1 + 1.$$

Relative to G_{012} , the same symbol is part of the declared seed data. These are different presentations, even when they generate isomorphic terminal arithmetic objects under later equivalence.

Definition 18 (Seed ambiguity). Seed ambiguity occurs when a construction has not declared whether it is using G_{01} , G_{012} , or another seed gauge. Until the gauge is declared, claims about primitivity, generation, minimality, and retained kernel remain underdetermined.

Postulate 1 (Seed no-leakage). A reconstruction may use seed-generated data only through operations legal in the declared seed gauge. If 2 is generated, it may be used after the generating operation is supplied. If 2 is primitive, that primitivity must be declared as part of the input data.

7 Clock coordinates

Definition 19 (Clock coordinate). A clock coordinate is an allowed operation axis. It is not automatically a parameter of ordinary time. It records how a system may be accessed, transformed, reconstructed, glued, or compared.

Definition 20 (θ -coordinate). The θ -coordinate is the derivation-access coordinate. It controls projection, probing, differentiation, deletion, coarse-graining, or observable extraction:

$$D_\theta : X \rightarrow Y.$$

It answers: what is legally exposed?

Definition 21 (α -coordinate). The α -coordinate is the integration or reconstruction coordinate. It controls gluing, completion, realization, transport, representative selection, or reconstruction:

$$I_\alpha : Y \times K \rightarrow X / \sim .$$

It answers: what is legally completed from the exposed data and retained kernel?

Definition 22 (Clock signature). For a problem P , the clock signature is

$$\text{Clk}(P) \subseteq \{\theta, \alpha\}.$$

The local/global diagnostic remains

$$\text{local} = \theta \oplus \alpha, \quad \text{global} = \theta \wedge \alpha, \quad \text{failure} = \text{wrong clock assignment}.$$

The clock-coordinate doctrine updates the earlier wording by making α the reconstruction coordinate axis and θ the access coordinate. A clocked argument must state which coordinate is active before interpreting a defect.

8 Clock-order residual

Definition 23 (Clock-order residual). Suppose both composites below are meaningful after mapping into a common residual carrier R :

$$D_\theta I_\alpha \quad \text{and} \quad I_\alpha D_\theta.$$

The clock-order residual is the declared comparison

$$\mathcal{E}_{\theta, \alpha}(X) = \text{Res}_R(D_\theta I_\alpha(X), I_\alpha D_\theta(X)).$$

When R is additive, one may write the shorthand

$$\mathcal{E}_{\theta,\alpha}(X) = D_\theta I_\alpha(X) - I_\alpha D_\theta(X).$$

The shorthand is admissible only after both terms have been typed in the same carrier.

Definition 24 (Flat order residual). The flat order residual is the same comparison with the flat integral:

$$\mathcal{E}_{\theta,\alpha}^0(X) = \text{Res}_R(D_\theta I_\alpha^0 D_\theta(X), D_\theta X).$$

It records the obstruction produced when visible data alone are reconstructed and reprojected.

Remark 3. The complete integral may be reversible by construction when the retained kernel is sufficient. The noncommutativity claim therefore belongs first to the clock-order residual of incomplete, flat, or incompatible reconstructions. A complete reconstruction still has to prove that the relevant diagram closes.

9 Retained kernel, admissibility, and reconstruction equivalence

Definition 25 (Retained-kernel class). For a Defective Calculus instance with $D_\theta : X \rightarrow Y$ and $I_\alpha : Y \times K \rightarrow X / \sim$, the retained-kernel class of X is

$$K_{\text{ret}}(X) = \{k \in K : I_\alpha(D_\theta X, k) \sim X \text{ and } \text{Adm}(X, k)\}.$$

Definition 26 (Admissible reconstruction). A reconstruction is admissible when:

1. the observable was produced by a declared D_θ ;
2. the reconstruction uses only declared I_α operations;
3. retained data lie in K_{ret} ;
4. no-leakage holds;
5. the result equals the target up to $\sim$, or the residual is explicitly recorded.

Definition 27 (No-leakage). No-leakage holds when every quantity used in a solution is either contained in the stated data or generated by a declared legal operation. Hidden data may be retained, but they may not be smuggled into visible certainty.

Postulate 2 (Retained-kernel discipline). A retained kernel is an admissible reconstruction class, not a rescue channel. If K is expanded to make a reconstruction succeed, the expansion must be typed, must preserve no-leakage, and must reject at least one nearby inadmissible reconstruction.

10 Thread extraction and glossary discipline

Definition 28 (Thread extraction record). A thread extraction record is a structured document of the form

$$T = (H, C, N, S, O, F, A, D),$$

where H is the source header, C is the core-claims list, N is the new-object list, S records structural changes, O records omissions, F records failure surfaces, A records action items, and D is the extraction digest.

A thread extraction should use the following fields.

Field	Required content
Source header	Thread title, date range, primary topic, framework objects involved, and status of the thread.
Core claims	Claim ID, claim, type, confidence, dependencies, conflicts, and formalization need.
New objects	Term, definition as used, stability, replacement wording if needed, related terms, and ambiguity risk.
Structural changes	What was added, removed, weakened, strengthened, clarified, or made more questionable.
Omissions	Undefined assumptions, proof obligations, missing examples, missing diagrams, and terminology needing standardization.
Failure surfaces	Weak point, reason weak, repair path, falsifier, and target document section.
Action items	Action, target document, priority, reason, input needed, and output expected.
Extraction digest	Minimal compressed restatement of what should survive into the next draft.

Definition 29 (Glossary entry). A glossary entry is a tuple

$$G_t = (t, d, s, r, u, a)$$

where t is the term, d is the definition, s is the stability status, r is replacement or deprecated wording, u is the upstream source, and a is the ambiguity risk.

Postulate 3 (Glossary no-drift rule). A term may not silently change meaning across documents. If a term changes, the later document must mark the change as a correction, restriction, extension, or replacement.

11 Failure surfaces

Code	Failure surface
A1	WDS is used as an ambient word without declaring its tuple data.
A2	COTG is treated as the whole WDS instead of a quotient or realization layer.
A3	Primal Calculus is stated as a universal inverse rather than a partial admissible factorization calculus.
A4	Scalar Calculus is used without a scale index, quotient map, retained kernel, and reconstruction equivalence.
A5	Reconstruction succeeds only because K was expanded into an arbitrary rescue channel.
A6	The seed gauge is not declared, so generated and primitive data are conflated.

A7	θ and α are used as decoration rather than typed operation coordinates.
A8	A clock-order residual is written as a commutator before both composites are typed in one residual carrier.
A9	Thread notes are imported as definitions without stability, dependencies, or failure surfaces.
A10	Glossary drift changes a term without declaring a correction or replacement.

12 Known unknowns and proof obligations

1. Specify the categories in which WDS tuples, quotient objects, sector decompositions, and reconstruction maps live.
2. Determine when θ and α can be derived from an object rather than assigned by the analyst.
3. Characterize which WDS sector decompositions are genuine direct sums, which are filtrations, and which are only presentation labels.
4. State the universal property, if any, of the original COTG skeleton $(A, B, C; E; K_2; \text{glue}_2)$.
5. Separate the seed-gauge problem from the arithmetic ladder problem: the former concerns primitive presentation, the latter concerns generated closure.
6. Define the minimal residual carrier needed for clock-order residuals without forcing all examples into an additive setting.
7. Give finite neighboring examples where retained-kernel admissibility succeeds in one case and fails in another.
8. Create a glossary versioning rule that is strict enough to prevent drift but not so strict that exploratory notes become impossible.

13 Unknown unknown controls

Unknown unknowns cannot be listed directly. This addendum therefore records control surfaces: every future document should state its carrier, equivalence, legal operations, quotient, retained kernel, residual carrier, admissibility rule, source status, and glossary changes. A missing item should be interpreted as an open defect, not as tacit permission.

14 Final compression

The addendum can be compressed as follows:

$$\text{DA}_{\text{global}} = (\text{seed gauge}, \text{WDS}, \text{Recon}, \text{DC}, \text{PC}, \text{SC}, \text{COTG}, K_{\text{ret}}, \text{Adm}, \text{thread discipline}).$$

The operational spine is

$$G \longrightarrow N \xrightarrow{\Delta} \Delta N \longrightarrow \text{sector data} \longrightarrow \text{COTG} \longrightarrow (\text{COTG}, K_{\text{ret}}) \xrightarrow{I} X / \sim .$$

The no-leakage spine is

visible quotient + admissible retained kernel $\neq$ arbitrary hidden rescue data.

The document-control spine is

thread extraction $\longrightarrow$ glossary entry $\longrightarrow$ typed draft $\longrightarrow$ reviewable failure surface.